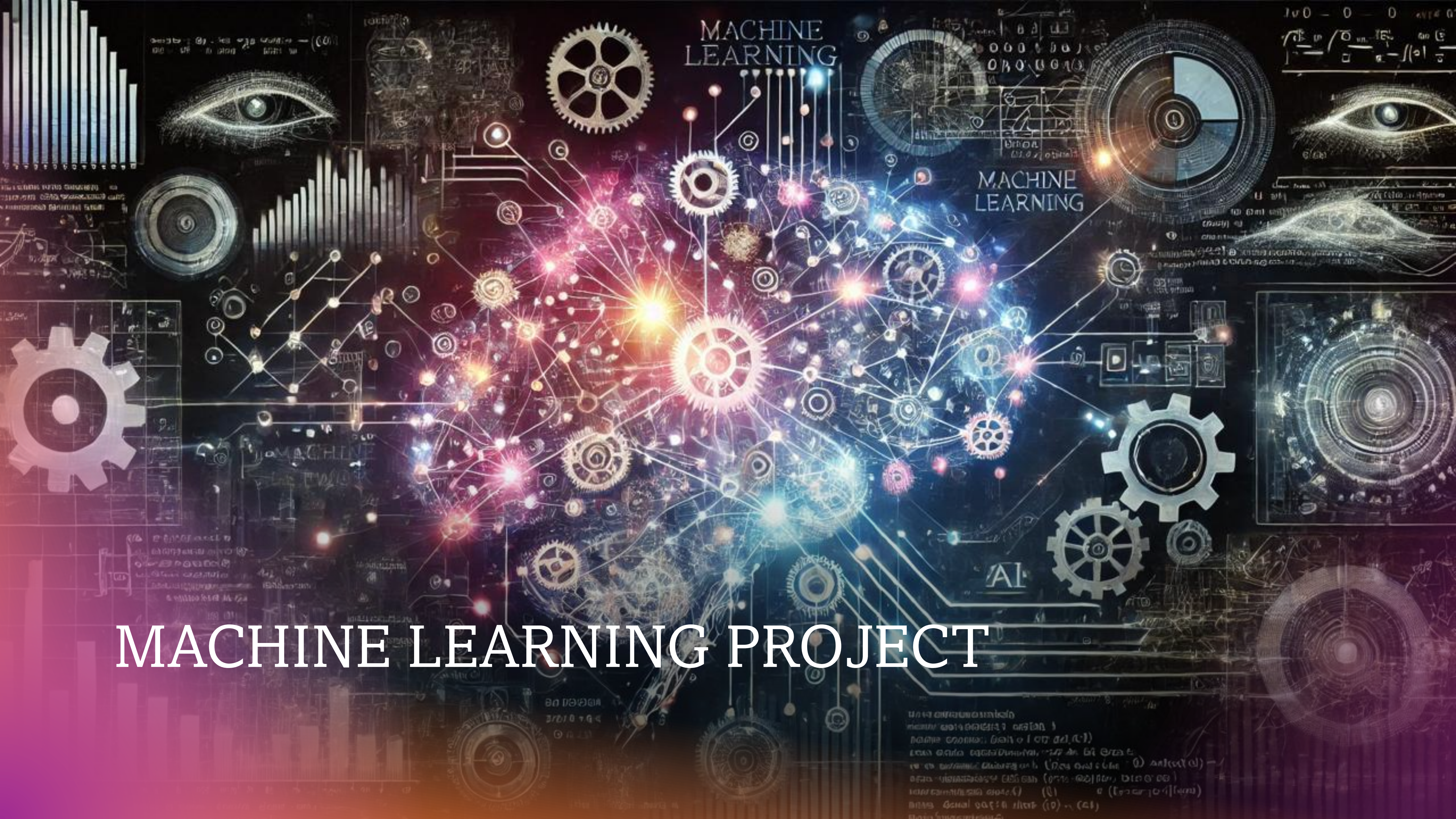


# MACHINE LEARNING PROJECT

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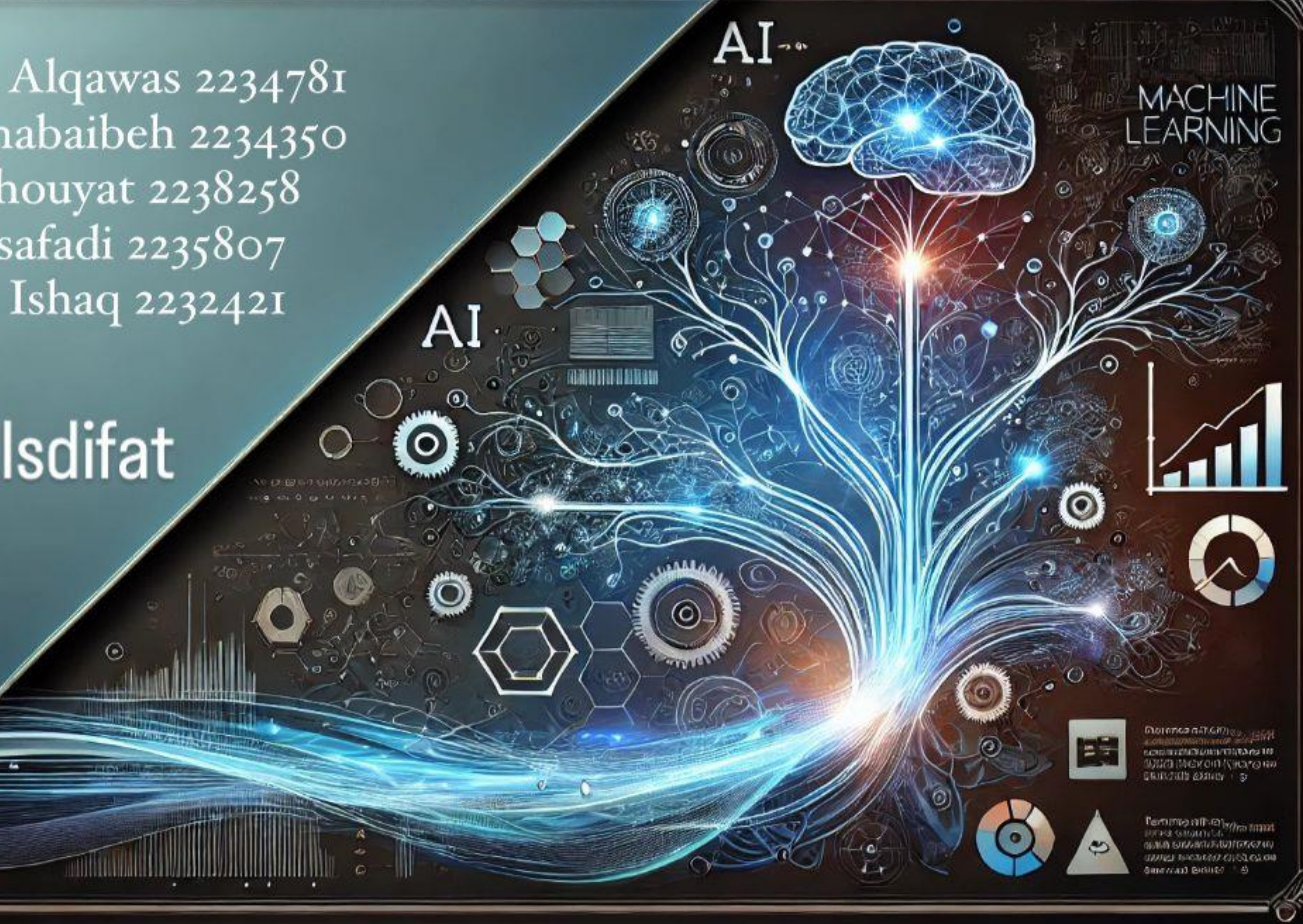


**Introduction to the Tasks Data Exploration and Visualization**

We will explore the"CO2 Emissions4".
This dataset provides detailed information about a variety of vehicles , including performance specifications, fuel consumption, and CO2 emissions. The columns include data such as vehicle type, engine size, number of cylinders , transmission type, and fuel type. It also records fuel consumption rates in the city,on highways, combined averages, as well as CO2 emissions and fuel
efficiency levels.

The dataset contains 7,385 records, covering various types and sizes of vehicles
. All columns are free of missing values, which simplifies handling and analysis
. Additionally, there are 5,692 duplicate rows, indicating the potential for data cleaning to improve analysis quality.

This dataset serves as a solid foundation for analyzing vehicle performance, environmental impact, and operational efficiency, allowing for the exploration of patterns in emissions and fuel consumption across different vehicle categories.

First 5 rows of the dataset:

| | Make | Model | 'Vehicle Class' | 'Engine Size(L)' | Cylinders | Transmission | 'Fuel Type' |
|---|---|---|---|---|---|---|---|
| 0 | PORSCHE | '911 GT3RS' | TWO-SEATER | 4.0 | 6 | AM7 | Z |
| 1 | BMW | M3 | COMPACT | 3.0 | 6 | M6 | Z |
| 2 | VOLKSWAGEN | 'GOLF SPORTWAGON TDI (modified)' | 'STATION WAGON - SMALL' | 2.0 | 4 | M6 | D |
| 3 | BMW | '335i xDRIVE SEDAN' | COMPACT | 3.0 | 6 | AS8 | Z |
| 4 | MASERATI | 'QUATTROPORTE GTS' | FULL-SIZE | 3.8 | 8 | AS8 | Z |

| 'Fuel Type' | 'Fuel Consumption City (L/100 km)' | 'Fuel Consumption Hwy (L/100 km)' | 'Fuel Consumption Comb (L/100 km)' | 'Fuel Consumption Comb (mpg)' | 'CO2 Emissions(g/km)' |
|---|---|---|---|---|---|
| Z | 16.3 | 11.8 | 14.3 | 20 | 334 |
| Z | 13.7 | 9.0 | 11.6 | 24 | 271 |
| D | 7.6 | 5.5 | 6.7 | 42 | 179 |
| Z | 11.9 | 7.8 | 10.0 | 28 | 230 |
| Z | 17.6 | 10.7 | 14.5 | 19 | 334 |

Dataset Information:

| | Column | Non-Null Count | Data Type |
|---|---|---|---|
| **Make** | Make | 7385 | object |
| **Model** | Model | 7385 | object |
| **'Vehicle Class'** | 'Vehicle Class' | 7385 | object |
| **'Engine Size(L)'** | 'Engine Size(L)' | 7385 | float64 |
| **Cylinders** | Cylinders | 7385 | int64 |
| **Transmission** | Transmission | 7385 | object |
| **'Fuel Type'** | 'Fuel Type' | 7385 | object |
| **'Fuel Consumption City (L/100 km)'** | 'Fuel Consumption City (L/100 km)' | 7385 | float64 |
| **'Fuel Consumption Hwy (L/100 km)'** | 'Fuel Consumption Hwy (L/100 km)' | 7385 | float64 |
| **'Fuel Consumption Comb (L/100 km)'** | 'Fuel Consumption Comb (L/100 km)' | 7385 | float64 |
| **'Fuel Consumption Comb (mpg)'** | 'Fuel Consumption Comb (mpg)' | 7385 | int64 |
| **'CO2 Emissions(g/km)'** | 'CO2 Emissions(g/km)' | 7385 | int64 |

```
Missing values per column:
```

| | Column | Missing Values |
|---|---|---|
| 0 | Make | 0 |
| 1 | Model | 0 |
| 2 | 'Vehicle Class' | 0 |
| 3 | 'Engine Size(L)' | 0 |
| 4 | Cylinders | 0 |
| 5 | Transmission | 0 |
| 6 | 'Fuel Type' | 0 |
| 7 | 'Fuel Consumption City (L/100 km)' | 0 |
| 8 | 'Fuel Consumption Hwy (L/100 km)' | 0 |
| 9 | 'Fuel Consumption Comb (L/100 km)' | 0 |
| 10 | 'Fuel Consumption Comb (mpg)' | 0 |
| 11 | 'CO2 Emissions(g/km)' | 0 |

Column Names:

| | Column Name |
|---|---|
| 0 | Make |
| 1 | Model |
| 2 | 'Vehicle Class' |
| 3 | 'Engine Size(L)' |
| 4 | Cylinders |
| 5 | Transmission |
| 6 | 'Fuel Type' |
| 7 | 'Fuel Consumption City (L/100 km)' |
| 8 | 'Fuel Consumption Hwy (L/100 km)' |
| 9 | 'Fuel Consumption Comb (L/100 km)' |
| 10 | 'Fuel Consumption Comb (mpg)' |
| 11 | 'CO2 Emissions(g/km)' |

Number of duplicated rows:

| | Metric | Count |
|---|---|---|
| 0 | Duplicated Rows | 5692 |

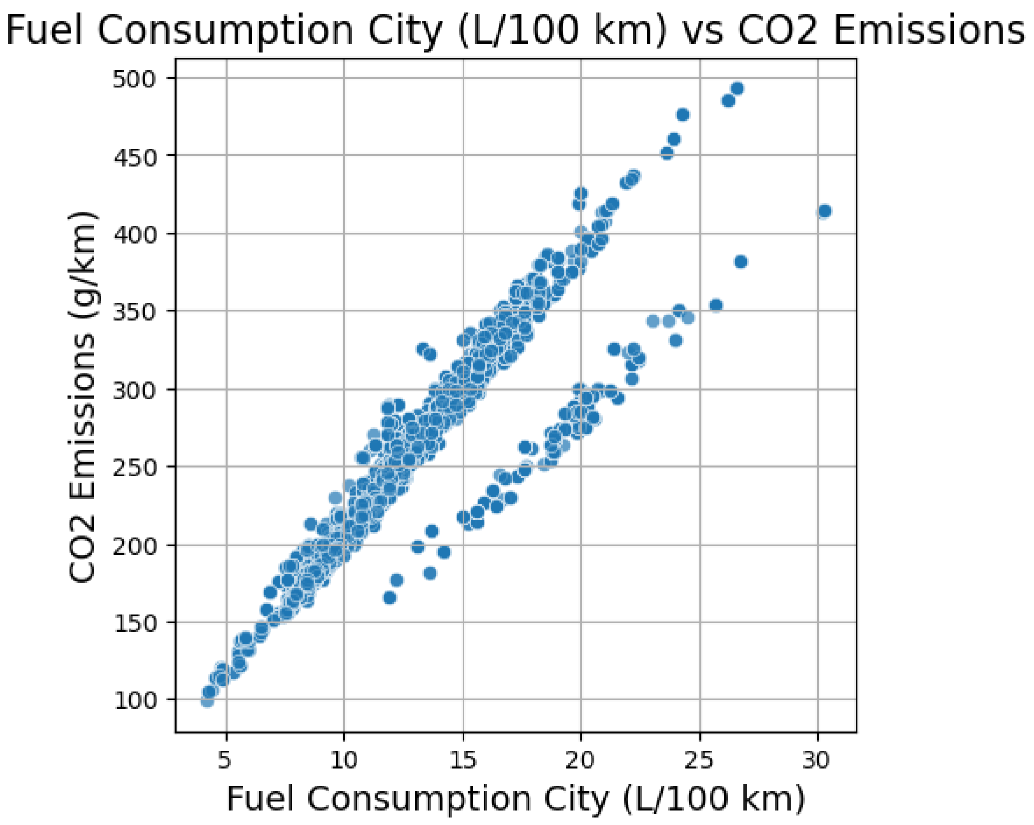
Dataset Dimensions:

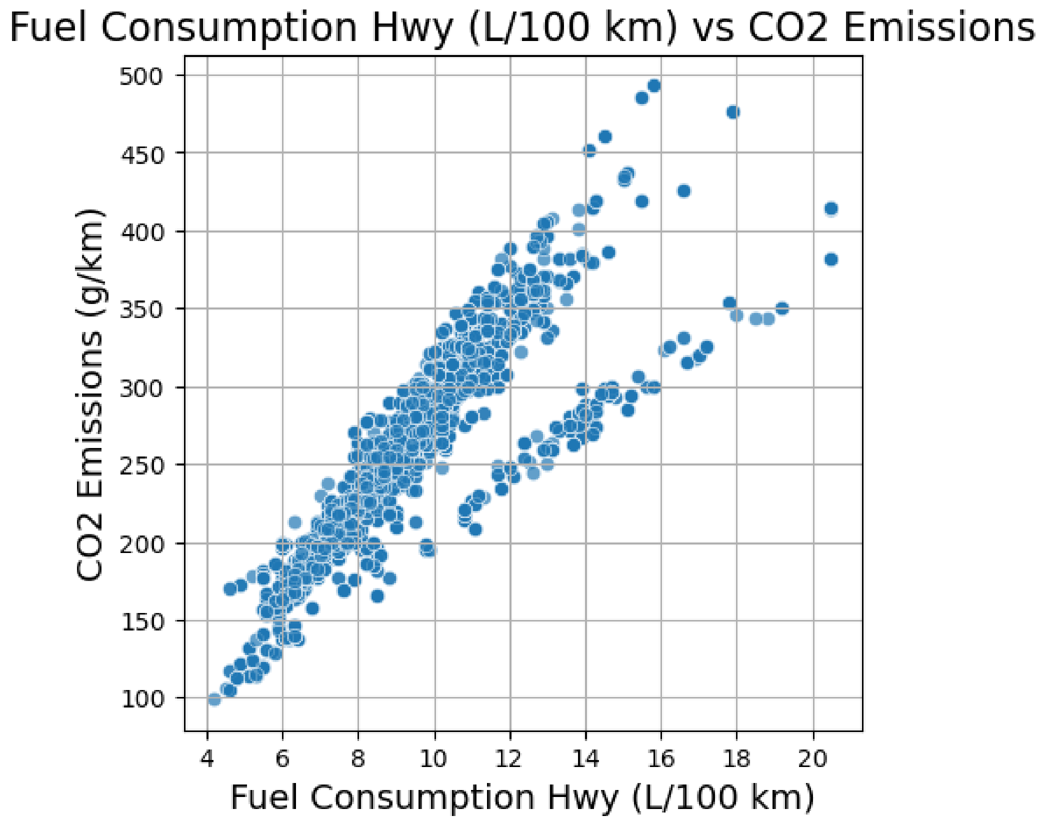
| | Metric | Count |
|---|---|---|
| 0 | Rows | 7385 |
| 1 | Columns | 12 |

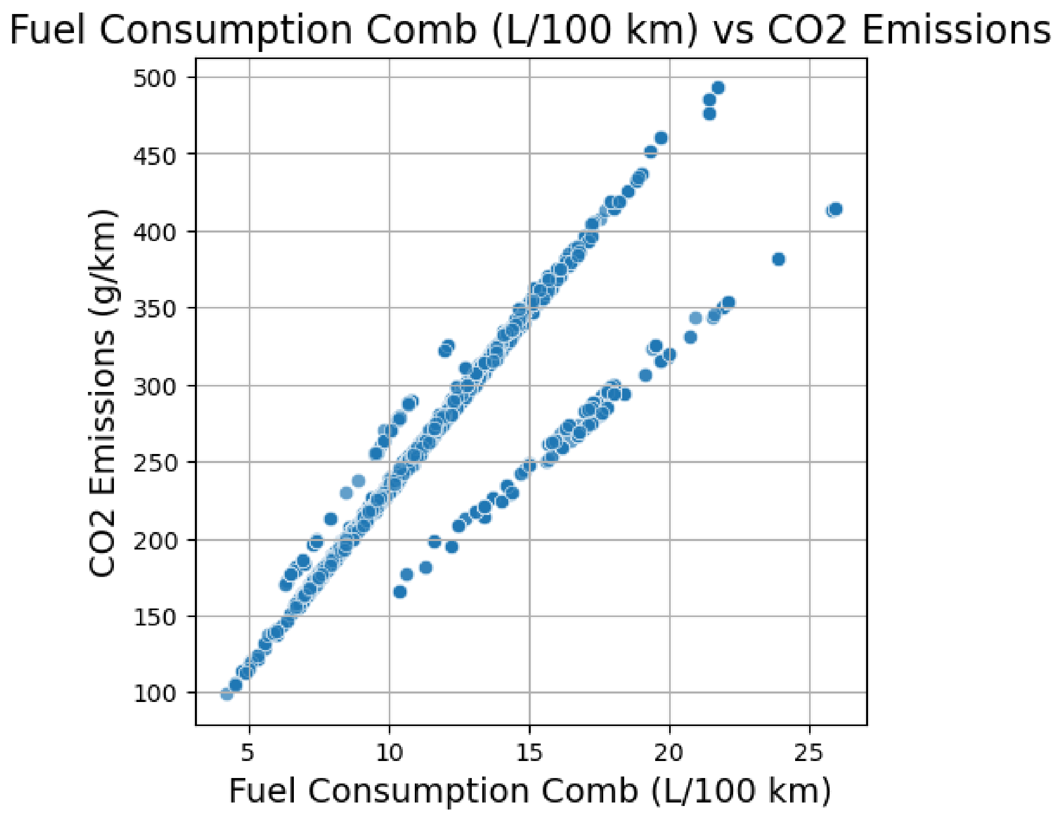
**Part 1**: Data Analysis and Visualization

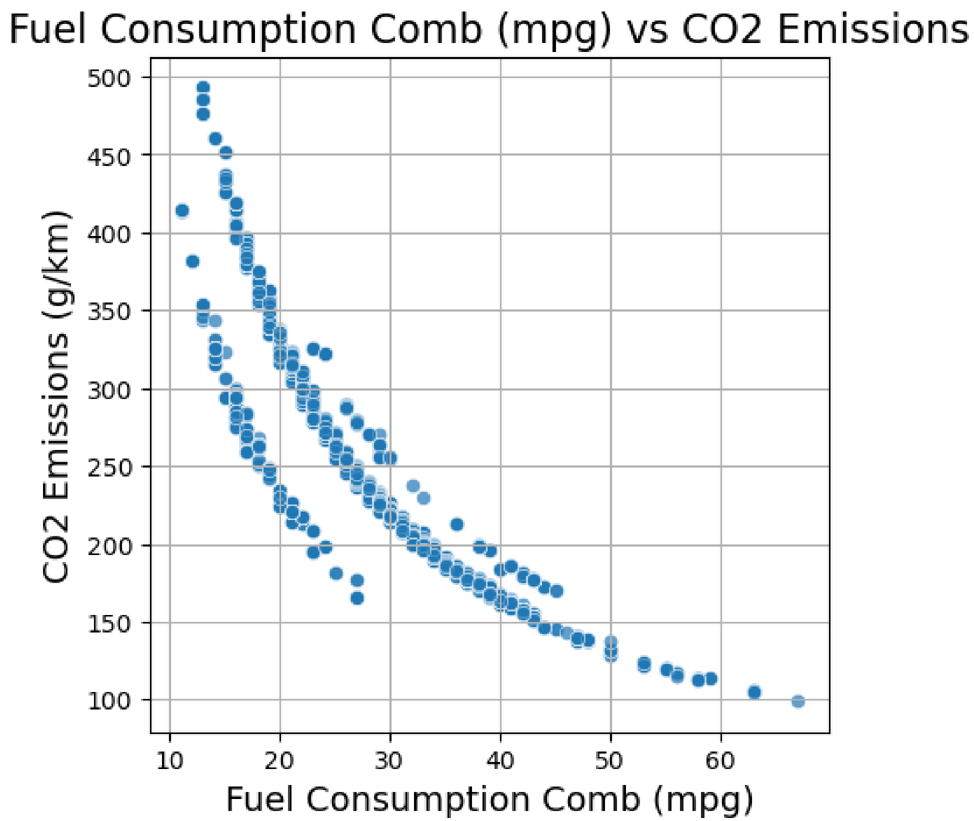
**Tasks :** 1. Visualization Tasks

Visualize the relationships between Engine Size, Fuel Consumption, and $CO_2$ Emissions using scatterplots.

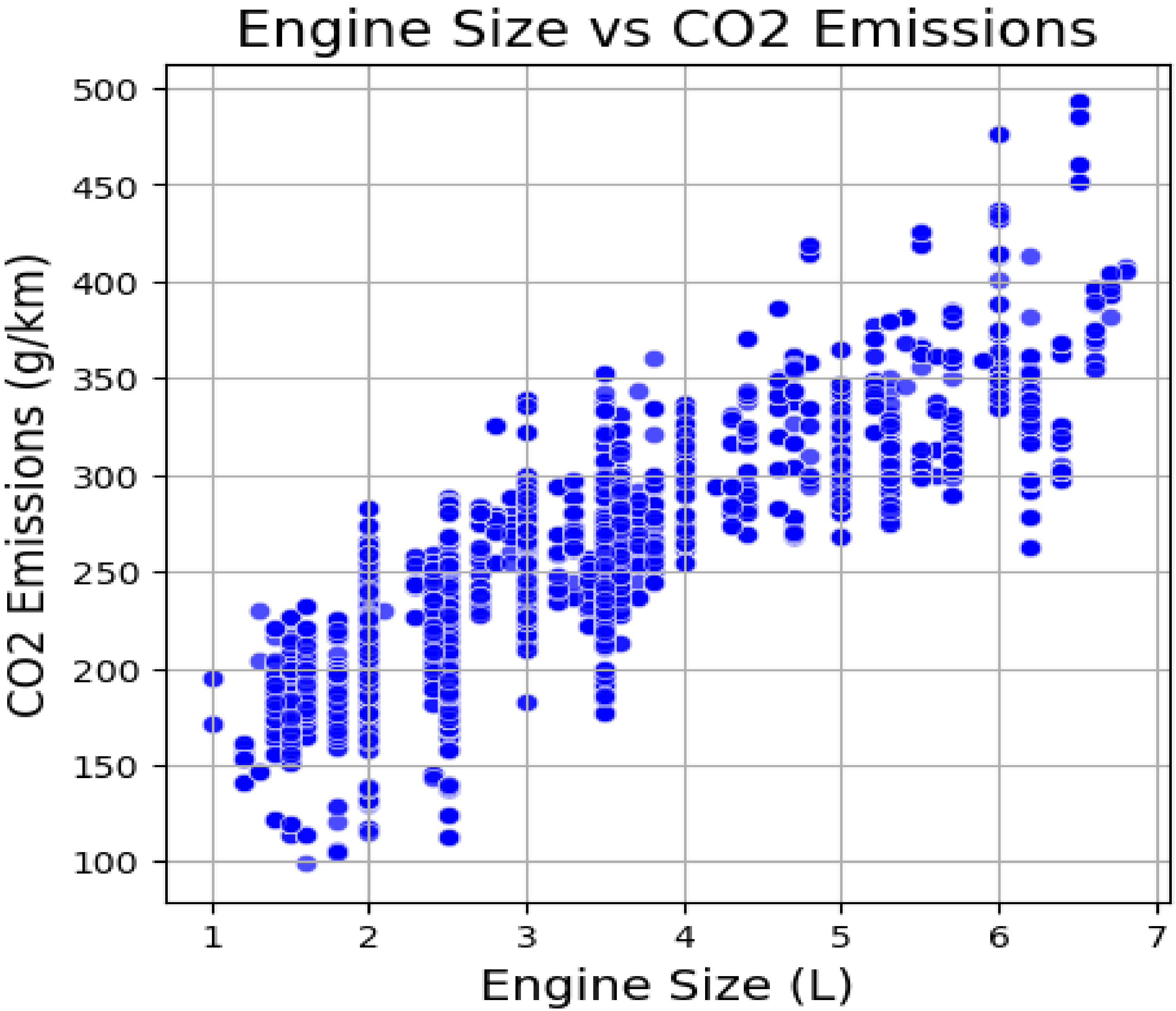


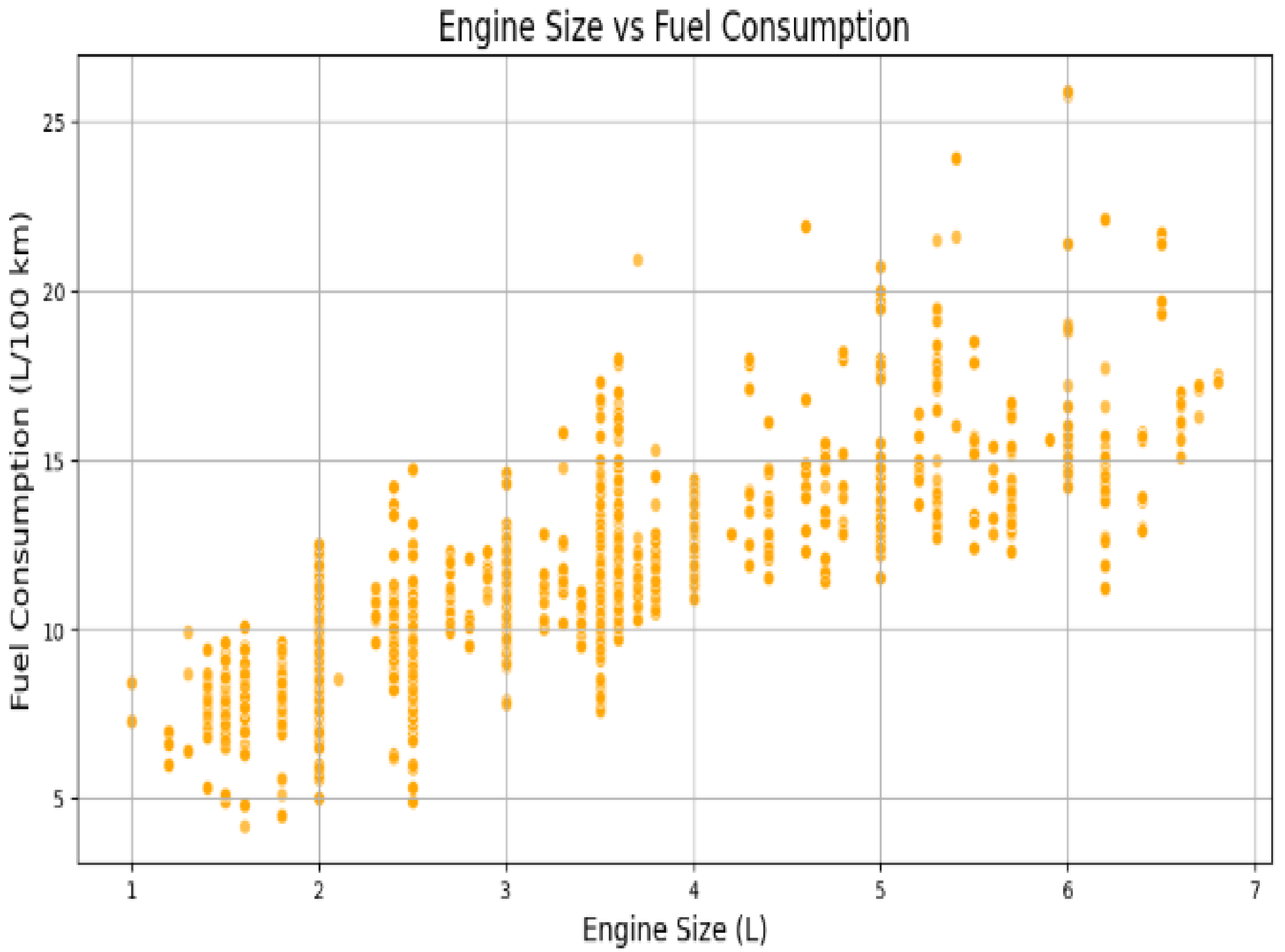

Fuel Consumption Comb (L/100 km) vs CO2 Emissions
CO2 Emissions (g/km)
Fuel Consumption Comb (L/100 km)
100
150
200
250
300
350
400
450
500
5
10
15
20
25


Fuel Consumption Comb (mpg) vs CO2 Emissions
CO2 Emissions (g/km)
Fuel Consumption Comb (mpg)
100
150
200
250
300
350
400
450
500
10
20
30
40
50
60

Engin size vs CO2 Emission :

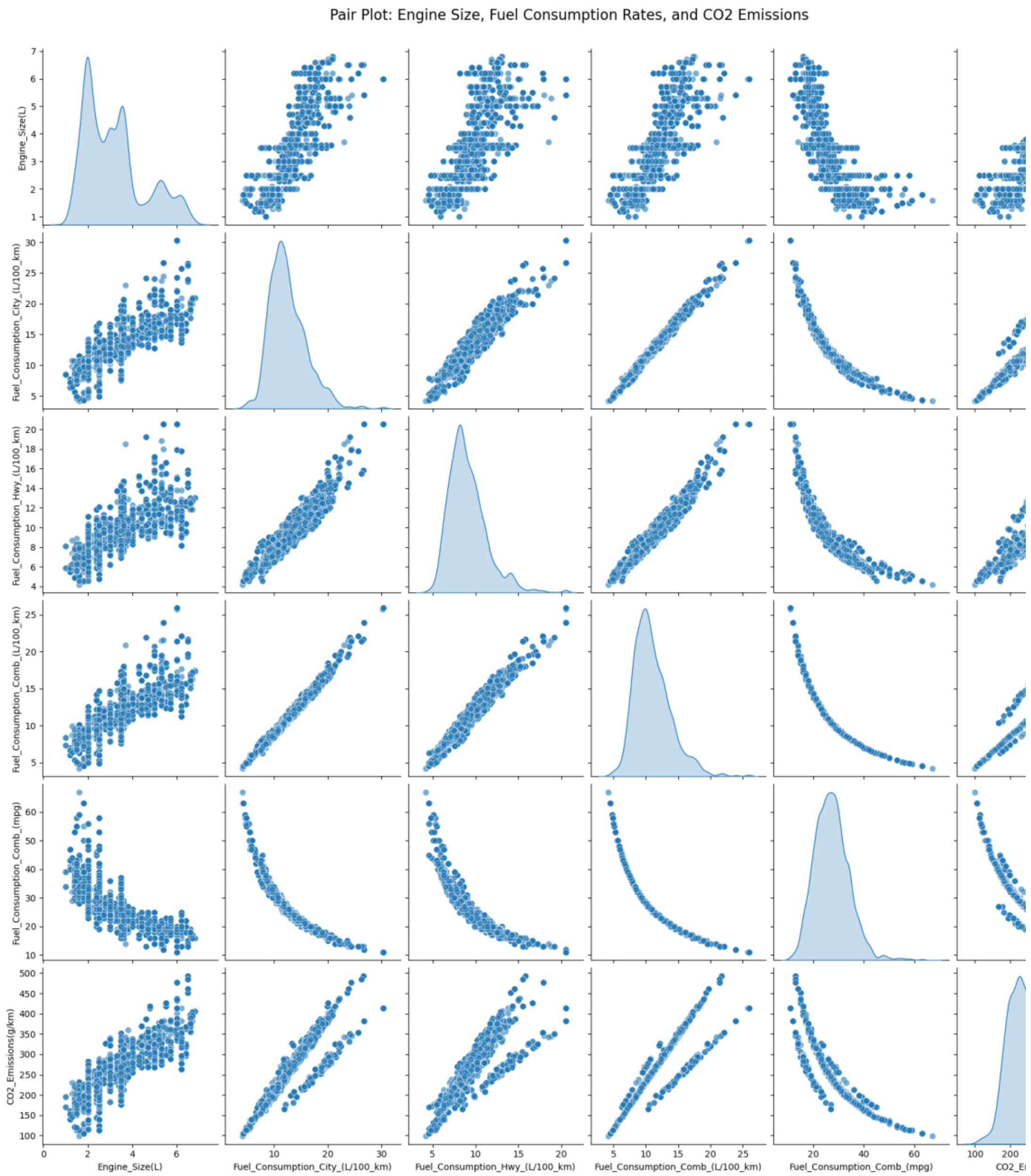


# Engine Size vs Fuel Consumption :

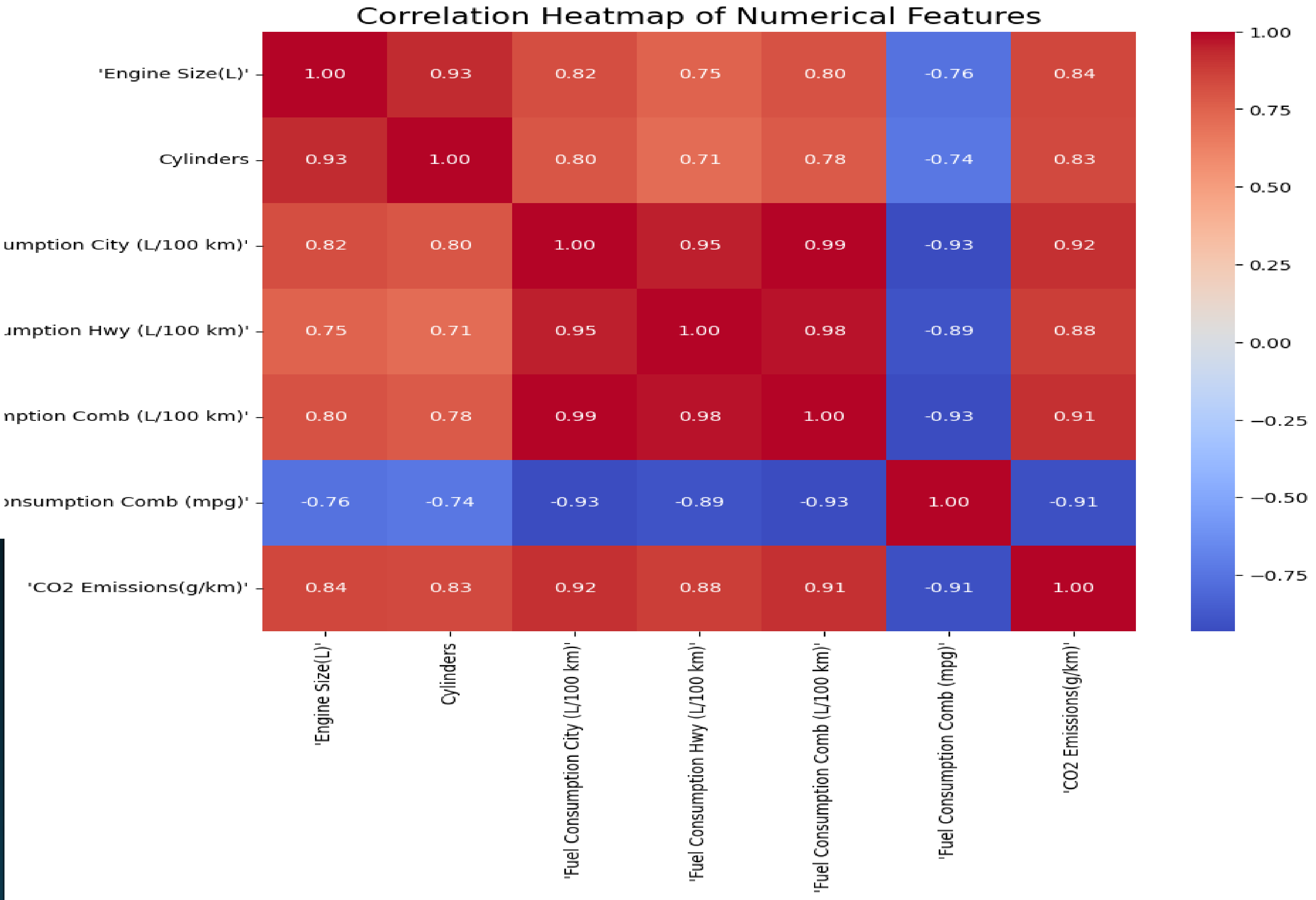


**Tasks : 1**. Visualization Tasks

Visualize the relationships between Engine Size, Fuel Consumption, and $CO_2$ Emissions using pair plots.



Create a heatmap of correlations to understand how numerical features are related to $CO_2$ Emissions.



## 2 . Feature Insights:

o Identify key features contributing to $CO_2$ emissions (e.g., does Fuel Type significantly impact emissions?).

o Provide a written summary of observed trends or patterns.

Feature Insights: Identifying Key Features Contributing to $CO_2$ Emissions

1 . Key Features Contributing to $CO_2$ Emissions Based on data analysis and feature correlations, the following are the primary factors contributing to $CO_2$ emissions:

Fuel Consumption Comb (L/100 km):

This is the most impactful feature on emissions. As fuel consumption (L/100 km) increases, $CO_2$ emissions increase proportionally. Vehicles with lower fuel efficiency, such as sports cars or those with large engines, record higher emissions.

Engine Size (L):

There is a strong positive relationship between engine size and emissions. Larger engines generate more power but also burn more fuel, leading to higher $CO_2$ emissions. Cylinders:

The number of cylinders is positively correlated with emissions. Engines with more cylinders are generally more powerful but less fuel-efficient. Fuel Consumption Comb (mpg):

There is a strong negative correlation with emissions. Vehicles with higher efficiency (more miles per gallon) produce significantly lower $CO_2$ emissions. Fuel Type:

Fuel type impacts emissions, with some types (like diesel) being more efficient than gasoline, resulting in lower emissions. Alternative fuels (such as electricity or natural gas) can drastically reduce emissions. 2. Observed Trends and Patterns Fuel Consumption and $CO_2$ Emissions:

A clear and strong correlation exists between fuel consumption (city, highway, and combined) and emissions. Higher fuel consumption directly results in higher emissions. Engine Size and Cylinders:
Vehicles with larger engines and more cylinders emit more $CO_2$, indicating that these characteristics are associated with higher fuel consumption. Fuel Type:
Fuel type plays an important role, but its impact depends on the overall efficiency of the vehicle. Electric or alternative-fuel vehicles record significantly lower emissions compared to diesel or gasoline. High-Efficiency Vehicles:
Vehicles with lower fuel consumption (more miles per gallon) and energy-saving technologies produce the least amount of $CO_2$ emissions.
Summary :
The primary features affecting $CO_2$ emissions are fuel consumption, engine size, and the number of cylinders. Vehicles with larger engines and higher cylinder counts consume more fuel and thus produce more emissions. Fuel type has a moderate impact, with alternative fuels outperforming conventional fuels in reducing emissions. The shift towards more efficient technologies, such as electric or hybrid engines, is essential for reducing emissions on a global scale.

## Part 2: Regression Analysis

Dataset: Use the same dataset as Part 1.

Tasks:

1 . Model Construction:

o Split the dataset into training (70%) and testing (30%) subsets. o Build a Multiple Linear Regression model to predict $CO_2$ Emissions using:

- Engine Size
- Fuel Consumption (Comb)
- Cylinders

```
Intercept: 24.017
Coefficients: 2.968, 18.272, 3.526

Model Performance:
Mean Squared Error (MSE): 176.402
R-squared (R²): 0.929

Predicted CO2 Emissions for the new record: 210.875
```

**Part 2:** Regression Analysis

**2 .** Feature Selection:

o Apply feature selection techniques (Forward technique) to refine the regression model.
o Rebuild the model using the selected features.

```
Selected Features: ['Fuel_Consumption_Comb_(L/100_km)', 'Cylinders', 'Engine_Si
ze(L)']

Final Model Summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:     CO2_Emissions(g/km)   R-squared:                       0.878
Model:                             OLS   Adj. R-squared:                  0.878
Method:                  Least Squares   F-statistic:                 1.237e+04
Date:                 Sun, 05 Jan 2025   Prob (F-statistic):               0.00
Time:                         00:26:12   Log-Likelihood:                -22830.
No. Observations:                 5169   AIC:                         4.567e+04
Df Residuals:                     5165   BIC:                         4.569e+04
Df Model:                            3
Covariance Type:             nonrobust
==============================================================================
====================
                                       coef    std err          t      P>|t|
[0.025      0.975]
------------------------------------------------------------------------------
const                               52.5800      1.295     40.604      0.000
50.041      55.119
Fuel_Consumption_Comb_(L/100_km)    12.9179      0.166     77.987      0.000
12.593      13.243
Cylinders                            7.5437      0.422     17.879      0.000
6.717       8.371
Engine_Size(L)                       4.5441      0.592      7.679      0.000
3.384       5.704
==============================================================================
Omnibus:                     1311.366   Durbin-Watson:                   1.954
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             4606.632
Skew:                          -1.246   Prob(JB):                         0.00
Kurtosis:                       6.896   Cond. No.                         64.4
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctl
y specified.

==============================================================================
```

**Part 2**: Regression Analysis

**3 .** Evaluation Metrics :

Evaluate model performance using:

R2 score.

Mean Squared Error (MSE).

```
Model Evaluation on Test Data:
Mean Squared Error (MSE): 427.475
R-squared (R²): 0.866
```

**4 . Insights:**

o Compare the initial and refined regression models based on evaluation metrics.

o Interpret the impact of key features on $CO_2$ emissions (e.g., how does an increase in engine size affect emissions?).

Comparison of Models:

Initial Model - $R^2$ Score: 0.866 | MSE: 427.475

Refined Model - $R^2$ Score: 0.866 | MSE: 427.475

Impact of Key Features on $CO_2$ Emissions:

An increase in Fuel_Consumption_Comb_(L/100_km) increases $CO_2$ emissions by 12.9 18 units.

An increase in Cylinders increases $CO_2$ emissions by 7.544 units.

An increase in Engine_Size(L) increases $CO_2$ emissions by 4.544 units.

Built a **Random Forest Regressor** model to predict target values using training data, then evaluated its performance using **Mean Squared Error (MSE)** and **R-squared ($R^2$)** metrics.

```python
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(n_estimators=100, max_depth=10, random_state=42)
rf.fit(X_train, y_train)


y_pred_rf = rf.predict(X_test)
mse_rf = mean_squared_error(y_test, y_pred_rf)
r2_rf = r2_score(y_test, y_pred_rf)

print("\nRandom Forest Model Performance:")
print(f"Mean Squared Error (MSE): {mse_rf:.3f}")
print(f"R-squared (R²): {r2_rf:.3f}")
```

```
Random Forest Model Performance:
Mean Squared Error (MSE): 43.781
R-squared (R²): 0.986
```

# Part 3: Classification
# A. Initial Data Exploration

First 5 rows of the dataset:

| | loan_id | ' no_of_dependents' | ' education' | ' self_employed' | ' income_annum' | ' loan_amount' | ' loan_term' | ' cibil_score' | ' residential_assets_value' | ' commercial_assets_value' | ' luxury_assets_value' | ' bank_asset_value' | loan_status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1103 | 1 | ' Graduate' | ' Yes' | 1600000 | 4900000 | 16 | 487 | 4300000 | 1100000 | 3900000 | 2000000 | ' Rejected |
| 1 | 2748 | 1 | ' Graduate' | ' Yes' | 9700000 | 25500000 | 18 | 741 | 21900000 | 6200000 | 34800000 | 4900000 | ' Approved |
| 2 | 3889 | 1 | ' Not Graduate' | ' No' | 9100000 | 23300000 | 18 | 792 | 800000 | 12600000 | 32100000 | 12400000 | ' Approved |
| 3 | 2921 | 1 | ' Not Graduate' | ' No' | 400000 | 1400000 | 4 | 302 | 200000 | 100000 | 1500000 | 500000 | ' Approved |
| 4 | 3589 | 4 | ' Not Graduate' | ' Yes' | 6900000 | 22000000 | 10 | 512 | 6100000 | 9100000 | 20600000 | 3900000 | ' Rejected |

Dataset Information:

| | Column | Non-Null Count | Data Type |
|---|---|---|---|
| **loan_id** | loan_id | 4269 | int64 |
| **' no_of_dependents'** | ' no_of_dependents' | 4269 | int64 |
| **' education'** | ' education' | 4269 | object |
| **' self_employed'** | ' self_employed' | 4269 | object |
| **' income_annum'** | ' income_annum' | 4269 | int64 |
| **' loan_amount'** | ' loan_amount' | 4269 | int64 |
| **' loan_term'** | ' loan_term' | 4269 | int64 |
| **' cibil_score'** | ' cibil_score' | 4269 | int64 |
| **' residential_assets_value'** | ' residential_assets_value' | 4269 | int64 |
| **' commercial_assets_value'** | ' commercial_assets_value' | 4269 | int64 |
| **' luxury_assets_value'** | ' luxury_assets_value' | 4269 | int64 |
| **' bank_asset_value'** | ' bank_asset_value' | 4269 | int64 |
| **' loan_status'** | ' loan_status' | 4269 | object |

Missing values per column:

| | Column | Missing Values |
|---|---|---|
| 0 | loan_id | 0 |
| 1 | ' no_of_dependents' | 0 |
| 2 | ' education' | 0 |
| 3 | ' self_employed' | 0 |
| 4 | ' income_annum' | 0 |
| 5 | ' loan_amount' | 0 |
| 6 | ' loan_term' | 0 |
| 7 | ' cibil_score' | 0 |
| 8 | ' residential_assets_value' | 0 |
| 9 | ' commercial_assets_value' | 0 |
| 10 | ' luxury_assets_value' | 0 |
| 11 | ' bank_asset_value' | 0 |
| 12 | ' loan_status' | 0 |

Column Names:

| | Column Name |
|---|---|
| 0 | loan_id |
| 1 | ' no_of_dependents' |
| 2 | ' education' |
| 3 | ' self_employed' |
| 4 | ' income_annum' |
| 5 | ' loan_amount' |
| 6 | ' loan_term' |
| 7 | ' cibil_score' |
| 8 | ' residential_assets_value' |
| 9 | ' commercial_assets_value' |
| 10 | ' luxury_assets_value' |
| 11 | ' bank_asset_value' |
| 12 | ' loan_status' |

Dataset Dimensions:

| | Metric | Count |
|---|---|---|
| 0 | Rows | 4269 |
| 1 | Columns | 13 |

```
Data types for columns :
loan_id                        int64
 no_of_dependents'             int64
' education'                   object
' self_employed'               object
' income_annum'                int64
' loan_amount'                 int64
' loan_term'                   int64
' cibil_score'                 int64
' residential_assets_value'    int64
' commercial_assets_value'     int64
' luxury_assets_value'         int64
' bank_asset_value'            int64
' loan_status'                 object
dtype: object
```

**Tasks:**
**1 .** Preprocess the dataset (handle missing values, encode categorical variables).

I handled the missing data, represented by zeros, by calculating the mean value and imputing it into these entries.

```
Imputed means:
no_of_dependents: 2.98
residential_assets_value: 7792397.24
commercial_assets_value: 5323549.24
bank_asset_value: 5123744.72
```

Sample of processed data:

| | loan_id | no_of_dependents | income_annum | loan_amount | loan_term | cibil_score |
|---|---|---|---|---|---|---|
| 0 | 1103 | 1.0 | 1600000 | 4900000 | 16 | 487 |
| 1 | 2748 | 1.0 | 9700000 | 25500000 | 18 | 741 |
| 2 | 3889 | 1.0 | 9100000 | 23300000 | 18 | 792 |
| 3 | 2921 | 1.0 | 400000 | 1400000 | 4 | 302 |
| 4 | 3589 | 4.0 | 6900000 | 22000000 | 10 | 512 |

| residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | education_' Not Graduate' | self_employed_' Yes' | loan_status_' Rejected' |
|---|---|---|---|---|---|---|
| 4300000.0 | 1100000.0 | 3900000 | 2000000.0 | 0 | 1 | 1 |
| 21900000.0 | 6200000.0 | 34800000 | 4900000.0 | 0 | 1 | 0 |
| 800000.0 | 12600000.0 | 32100000 | 12400000.0 | 1 | 0 | 0 |
| 200000.0 | 100000.0 | 1500000 | 500000.0 | 1 | 0 | 0 |
| 6100000.0 | 9100000.0 | 20600000 | 3900000.0 | 1 | 1 | 1 |

## **2** . Apply various classification

## **First :**

DecisionTree

```
Decision Tree Classifier:
Accuracy: 0.99
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       794
           1       0.99      0.99      0.99       487

    accuracy                           0.99      1281
   macro avg       0.99      0.99      0.99      1281
weighted avg       0.99      0.99      0.99      1281
```

**second :**
LogisticRegression

```
Logistic Regression:
Accuracy: 0.81
              precision    recall  f1-score   support

           0       0.81      0.90      0.85       794
           1       0.80      0.67      0.73       487

    accuracy                           0.81      1281
   macro avg       0.81      0.78      0.79      1281
weighted avg       0.81      0.81      0.81      1281
```

# **Third** : Naïve Bayes

```
Naïve Bayes:
Accuracy: 0.75
              precision    recall  f1-score   support

           0       0.73      0.96      0.83       794
           1       0.86      0.42      0.57       487

    accuracy                           0.75      1281
   macro avg       0.80      0.69      0.70      1281
weighted avg       0.78      0.75      0.73      1281
```

# **Fourth** : RandomForestClassifier (Ensamble)

```
Random Forest Classifier:
Accuracy: 0.99
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       794
           1       0.99      0.99      0.99       487

    accuracy                           0.99      1281
   macro avg       0.99      0.99      0.99      1281
weighted avg       0.99      0.99      0.99      1281
```

# **last classifier** : SVM with StandardScaler

Support Vector Machine (with Feature Scaling):
Accuracy: 0.90

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.95 | 0.89 | 0.92 | 794 |
| 1 | 0.84 | 0.93 | 0.88 | 487 |
| accuracy | | | 0.90 | 1281 |
| macro avg | 0.89 | 0.91 | 0.90 | 1281 |
| weighted avg | 0.91 | 0.90 | 0.90 | 1281 |

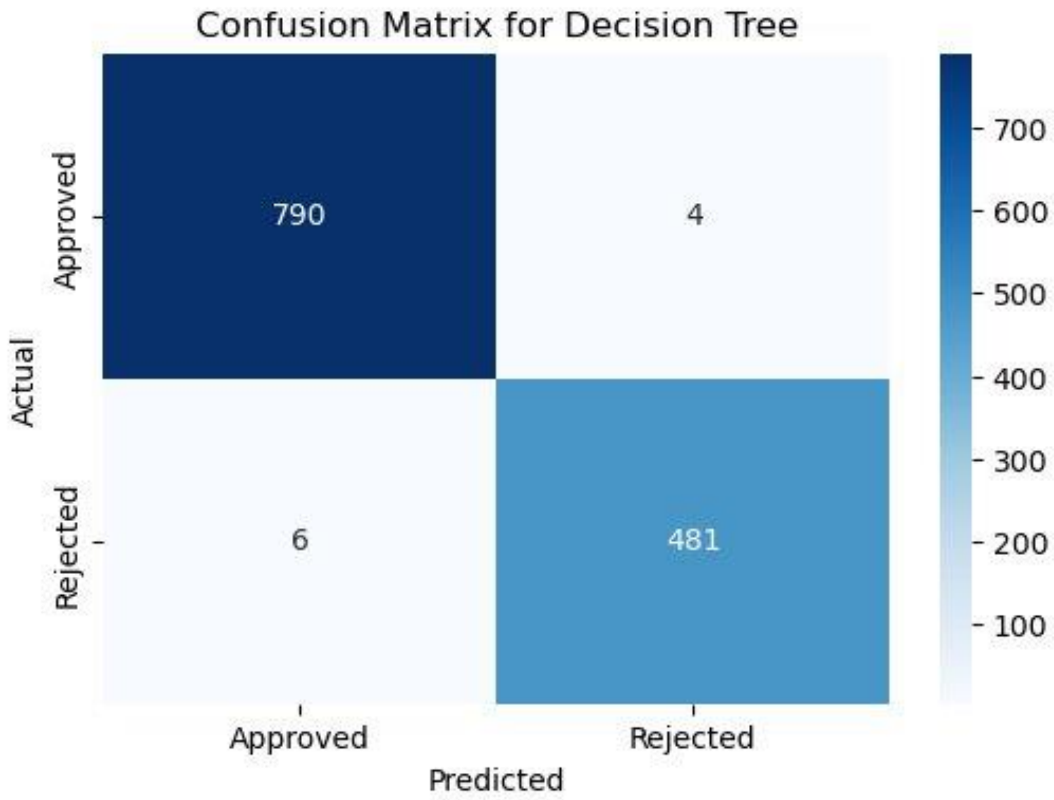
**3-** Evaluate models using confusion matrices and performance metrics (Accuracy, Precision, Recall, F1-Score, AUC).

```
=== Performance Metrics for Decision Tree ===
Accuracy: 0.99
Precision: 0.99
Recall: 0.99
F1-Score: 0.99

Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       794
           1       0.99      0.99      0.99       487

    accuracy                           0.99      1281
   macro avg       0.99      0.99      0.99      1281
weighted avg       0.99      0.99      0.99      1281
```

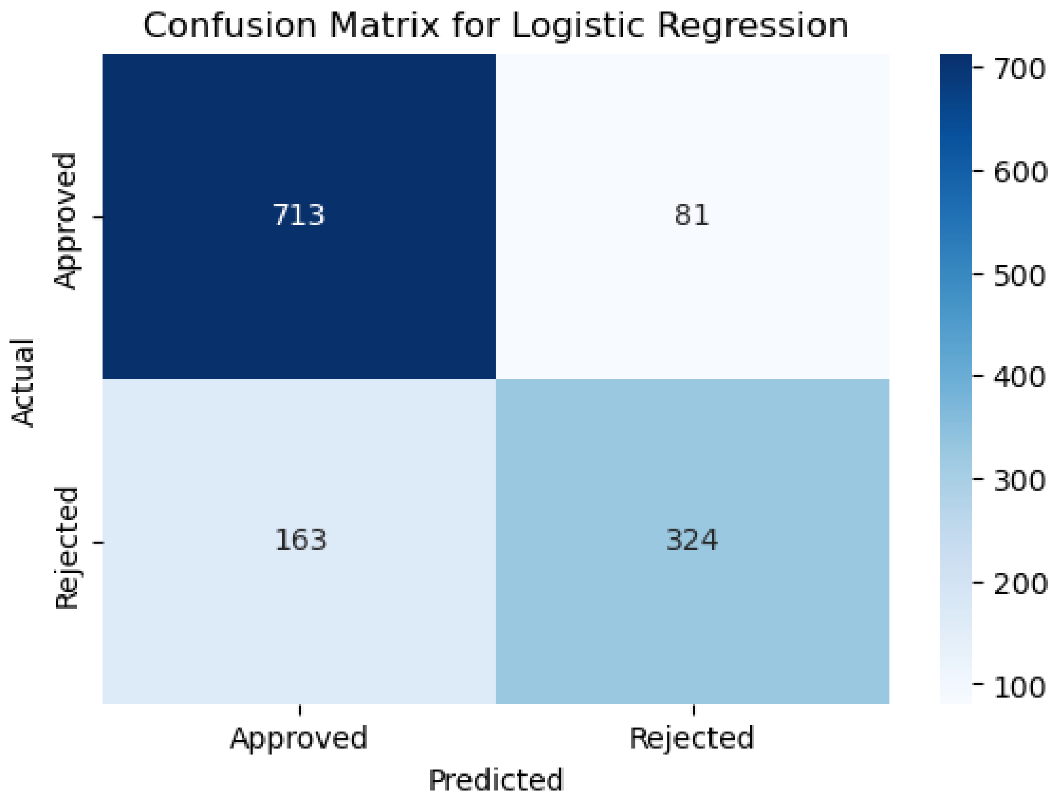


```
=== Performance Metrics for Logistic Regression ===
Accuracy: 0.81
Precision: 0.80
Recall: 0.67
F1-Score: 0.73

Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.90      0.85       794
           1       0.80      0.67      0.73       487

    accuracy                           0.81      1281
   macro avg       0.81      0.78      0.79      1281
weighted avg       0.81      0.81      0.81      1281
```

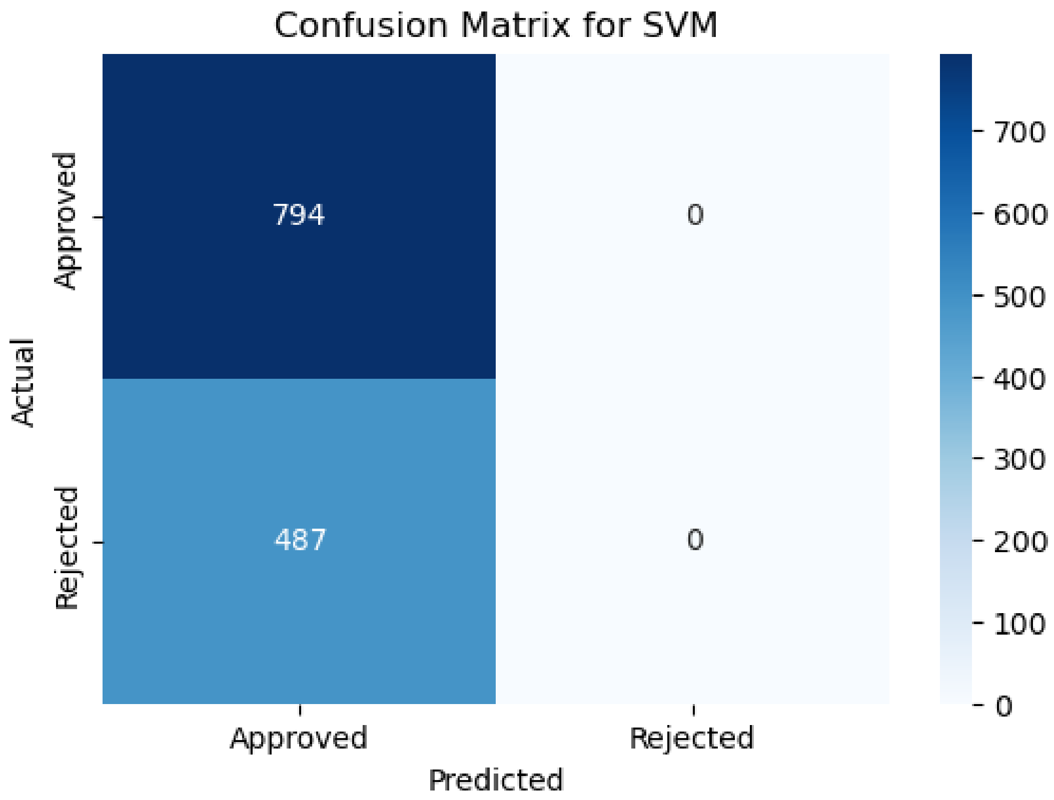


```
=== Performance Metrics for SVM ===
Accuracy: 0.62
Precision: 0.00
Recall: 0.00
F1-Score: 0.00

Classification Report:
              precision    recall  f1-score   support

           0       0.62      1.00      0.77       794
           1       0.00      0.00      0.00       487

    accuracy                           0.62      1281
   macro avg       0.31      0.50      0.38      1281
weighted avg       0.38      0.62      0.47      1281
```

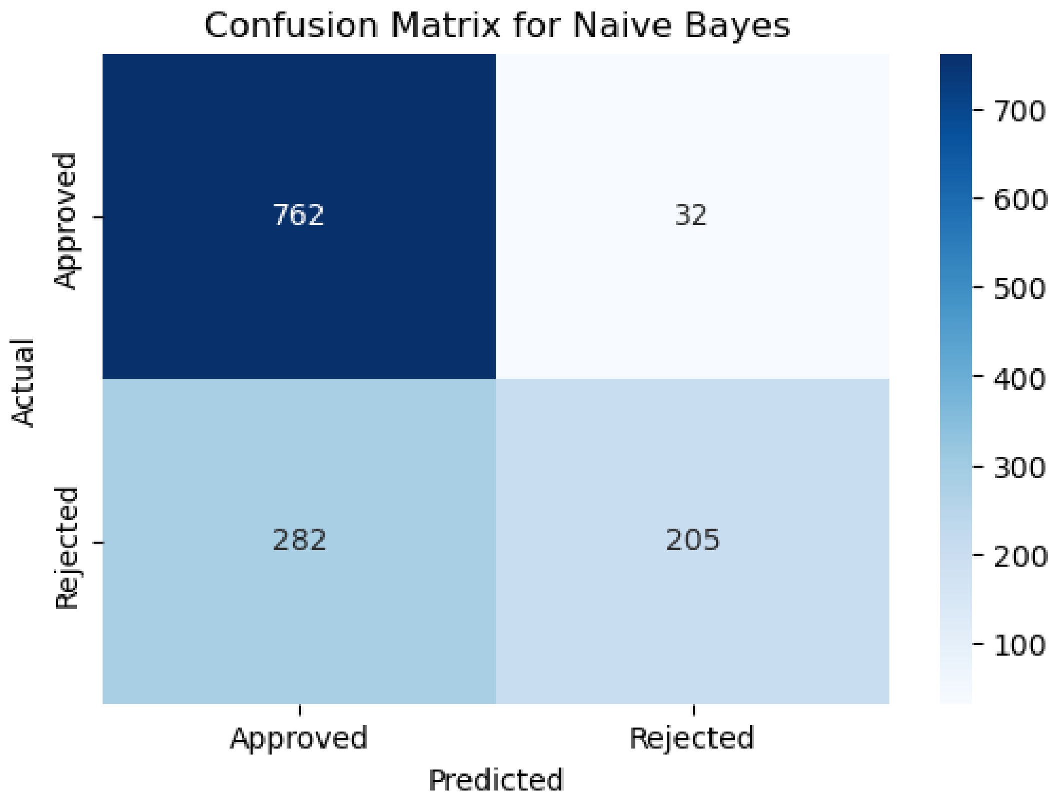


```
=== Performance Metrics for Naive Bayes ===
Accuracy: 0.75
Precision: 0.86
Recall: 0.42
F1-Score: 0.57

Classification Report:
              precision    recall  f1-score   support

           0       0.73      0.96      0.83       794
           1       0.86      0.42      0.57       487

    accuracy                           0.75      1281
   macro avg       0.80      0.69      0.70      1281
weighted avg       0.78      0.75      0.73      1281
```

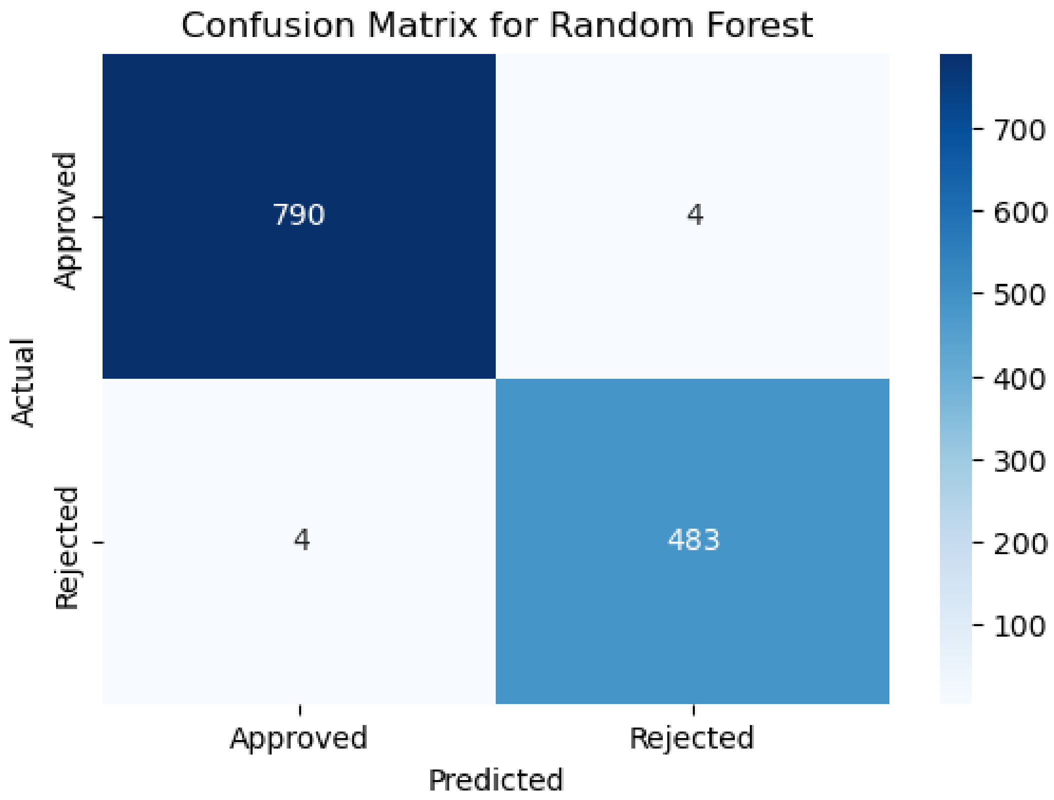


```
=== Performance Metrics for Random Forest ===
Accuracy: 0.99
Precision: 0.99
Recall: 0.99
F1-Score: 0.99

Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       794
           1       0.99      0.99      0.99       487

    accuracy                           0.99      1281
   macro avg       0.99      0.99      0.99      1281
weighted avg       0.99      0.99      0.99      1281
```



# 3 . Compare hold-out validation (70/30 split) with 10-fold cross validation.

```
=== Holdout Validation ===
Holdout Accuracy: 0.992

=== 10-Fold Cross Validation ===
Cross-Validation Accuracy Scores: 0.995 / 1.000 / 0.995 / 0.991 / 1.000 / 0.998
/ 0.988 / 0.998 / 0.993 / 0.993
Mean Accuracy: 0.995
Standard Deviation: 0.004
```


## 4 . Identify the classifier (Classification model) that the group has selected (the type, it's performance, reasons for selection).

The type : Random Forest(Ensemble)

it's performance : 99%

Reasons for selection : After building various models, it was found that Decision Tree and Random Forest provided the best accuracy. However, given the nature of the data, which may suffer from feature overlap, this could lead to overfitting in the case of the Decision Tree. Therefore, we chose Random Forest as the best classifier since it delivers optimal performance with medium to large datasets. Our dataset, which is considered medium-sized, achieved a very high accuracy of 99%, making Random Forest an excellent choice.

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